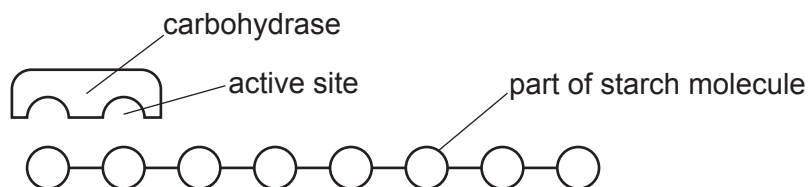


8. **Image 8.1** shows part of a starch molecule and carbohydrase.

Image 8.1



- (a) (i) State the name of the model used to explain enzyme action. [1]

.....

- (ii) All enzymes are proteins.
State what determines the shape of the active site of an enzyme. [1]

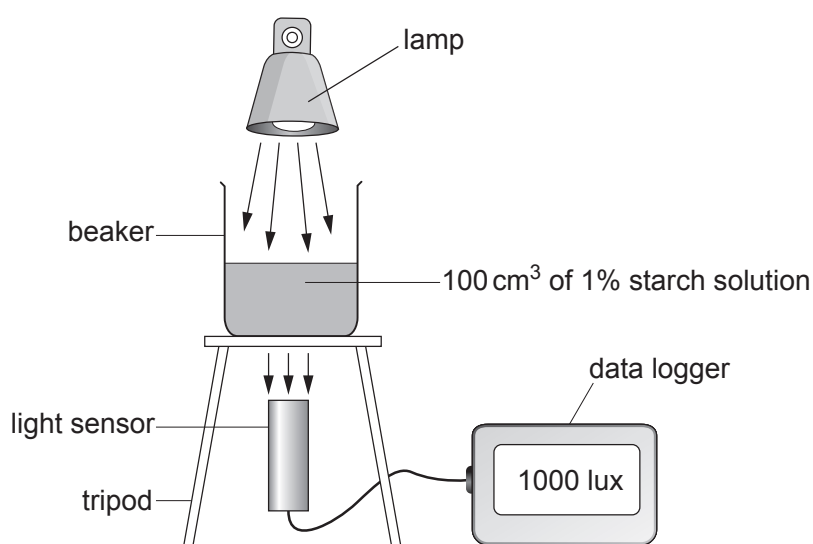
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- (b) An investigation was carried out to study the rate of digestion of starch by carbohydrase, using a light sensor linked to a data logger. The light sensor detects the intensity of light passing through a solution.

Image 8.2 shows how the investigation was set up.

Image 8.2

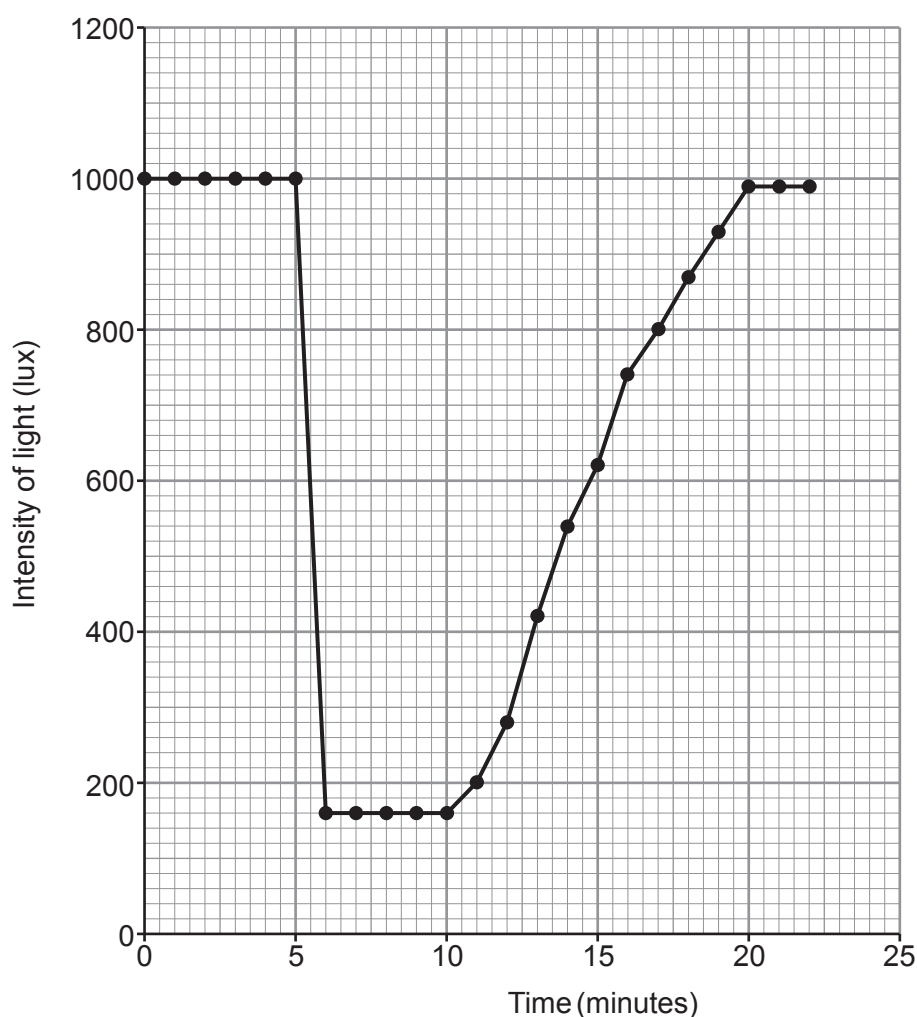


The following method was used.

- A lamp was placed above a beaker containing 100 cm³ of 1% starch solution.
- A light sensor was placed underneath the beaker and a data logger began recording the light intensity below the beaker.
- At 5 minutes, a few drops of iodine were added to the beaker.
- At 10 minutes, 20 cm³ of 10% carbohydrase solution was added.

The results of the investigation are shown in **Graph 8.3**.

Graph 8.3



(i) Explain the results between 5 and 6 minutes.

[2]

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.....

.....



- (ii) Calculate the mean rate of reaction between 10 and 15 minutes, using the following formula. [3]

$$\text{Mean rate of reaction} = \frac{\text{change in light intensity}}{\text{time}}$$

Mean rate of reaction = lux/min

- (iii) Explain the change in light intensity between 10 and 20 minutes. [3]

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.....

.....

- (iv) State **two** controlled variables in this experiment. [2]

1.
2.

- (v) **Draw a line** on **Graph 8.3** to predict the results if the carbohydrase had been boiled and then cooled before being added at 10 minutes. [1]

